MXET 375 Exam Formula Sheet

Translational Inertia Element (Mass):

$m\ddot{x}(t) = F_m(t) \;=\; \Sigma F$

Translational Stiffness Element (Spring):

$F_k = k[x_2(t) - x_1(t)]$

Translational Damping Element (Damper):

$F_b(t) = b[\dot{x_2}(t) - \dot{x_1}(t)]$

Rotational Inertia Element:

$j\ddot{\theta} = T_j(t), \theta_1$

Rotational Stiffness Element (Spring):

$T_k(t) = k[\theta_2(t) - \theta_1(t)]$

Rotational Damping Element (Damper):
$T_b(t) = b\left[\dot{\theta}(t) - \dot{\theta_1}(t)\right]$

Ideal Lever:

$$F_1 L_1 = F_2 L_2$$

Gear Ratio:

$\frac{r_2}{r_1} = \frac{n_2}{n_1} = N$

Equating Mesh Velocities:

$V_{mesh} = r_1\omega_1 = r_2\omega_2$

Equating Input Power = Output Power:

$T_1\omega_1 = T_2\omega_2$

$\frac{T_1}{T_2} = \frac{\omega_2}{\omega_1}$

$GR = N = \frac{n_2}{n_1} = \frac{r_2}{r_1} = \frac{D_2}{D_1} = \frac{T_2}{T_1} = \frac{\omega_1}{\omega_2}$

Pendulum:

$I\ddot{\theta} = T_c(t) = \Sigma F, \theta_1$

State Space Representation:

One First Order ODE $\xrightarrow{\text{State Space Model}}$ 1 State Variable

One Second Order ODE $\xrightarrow{\text{State Space Model}}$ 2 State Variables

Two Second Order ODE $\xrightarrow{\text{State Space Model}}$ 4 State Variables

1. System Order & Variables
   a. SO? SV? Inputs? Outputs?
2. EOM then Simplify, isolate xdot on left side. Sort by order.
3. Convert to First Order ODE
   a. $x_1 = x$ (displacement)
   b. $x_2 = \dot{x}$ (velocity)
   c. d/dt 'em

$$x_1 = x \rightarrow \frac{d}{dt} \rightarrow \dot{x_1} = \dot{x} = x_2$$

   d. replace $\dot{x}$ with step a-b variable
   e. Assemble State Space matrices

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Where A is state matrix
Where B is input matrix
Where C is output matrix
Where D is direct-link matrix

Modeling of Electrical Systems:

1. Label / sum currents
2. Name state variables
3. In/Out
4. Label node, KCL Σi=0, ohm's 'em
5. define mesh & loop dir, KVL ΣV=0
6. Out = In (derivative order sort)

Resistor:
Voltage Drop: $v \;=\; iR$
Power Dissipated: $P_{\text{watt}} = -vi = -Ri^2$

Capacitor:
Charge-Voltage: $q \;=\; Cv$
Current: $i = \frac{dq}{dt}$ or $i_{C/n} = C_n \frac{dv_{C_n}}{dt}$
Energy Stored: $E = \frac{1}{2}Cv^2$
Power Dissipated: $P = Cv\frac{dv}{dt}$

Inductor:
Flux-Current: $\lambda = \; Li$
Voltage: $v = \frac{d\lambda}{dt}$ or $V_{L/n} = L_n \frac{d_i(t)}{dt}$
Energy Stored: $E = \frac{1}{2}Li^2$
Power Dissipated: $P = Li\frac{di}{dt}$

Inputs: i_{in} and/or V_{in} only
Outputs: i_{L_n} and/or V_{C_n} only

Combined Mechanical Systems:

0. Simple 1 whole FBD
1. Inputs, Outputs
2. Separate 1,2,etc FBD

Pendulum: (↑V(y),→H(x)) (yG, xG)
Body: (↓V, ←H),(u,x,xddot)

3.1. EoM

$$M\ddot{x} =/ \Sigma F_{\text{pendulum}} \;=\; u - H$$

$$I\ddot{\theta} = \Sigma T \;=/\; VL\sin\theta - HL\cos\theta + \cancel{\mathrm{mgcos}(\theta)}$$

3.2. FBD EoMs

$$m\ddot{x_G} = \Sigma F_x \;/=\; \mathrm{H} \;=/m\left(\ddot{x} + l\cos\theta\ddot{\theta} - l\sin\theta\,\theta^2\right) = H$$

$$m\ddot{y_G} = \Sigma F_y \;/=\; \mathrm{V - mg} \;=/m\left(-l\sin\theta\ddot{\theta} - l\cos\theta\,\theta^2\right) = V - mg$$

3.3 EoM cont

$$x_G = x + L\sin\theta$$

$$y_G = L\cos\theta \;\; (if \;\uparrow, (-\downarrow))$$

$$\dot{x_G} = \dot{x} + L(\cos\theta)\dot{\theta}$$

$$\dot{y_G} = -L(\sin\theta)$$

$$\ddot{x}_G = \ddot{x} + L[-\sin\theta\theta^2 + \cos\theta\ddot{\theta}]$$

$$\ddot{y}_G = -L[\cos\theta\dot{\theta}^2] + \sin\theta\ddot{\theta}]$$

3.3.2 sub to 3.2=/equation (mxg)

3.4: Sub =/3.2 H,V to 3.2

$$(M+m)\ddot{x} + mL\cos(\theta\ddot{\theta}) - mL\sin\dot{\theta}^2 = u$$

$$(I + mL^2)\ddot{\theta} - mgL\sin\theta + mL\cos\ddot{\theta} = 0$$

4. N[th] order [linear/non-linear] ODE

Misc:

$$i_L - i_R - i_c = 0$$
$$i_L - \frac{V}{R} - c\frac{dvc}{dt}$$

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3 Cont

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} p(t) \\ q(t) \\ \dot{p}(t) \\ \dot{q}(t) \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ a_{13} & a_{23} & a_{33} & a_{34} \\ a_{14} & a_{24} & a_{35} & a_{34} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ u_3 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$y = x_1$